

MOHAMED JUBAIR K

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Professional Summary

AI & Data Science Engineer with 3+ years of experience in deep learning, full-stack development, and scalable AI systems. Proficient in PyTorch, TensorFlow, React.js, and cloud-native architectures. Built 10+ production-ready AI applications with measurable business impact. Passionate about Machine Learning, NLP, and building intelligent systems.

Education

C.K. College of Engineering and Technology, Cuddalore

2022 – 2026

B.Tech in Artificial Intelligence & Data Science

CGPA: 8.5/10

Technical Skills

Programming Languages Python, JavaScript, SQL, Java, C++

AI & Machine Learning PyTorch, TensorFlow, Scikit-learn, Hugging Face, Transformers

Deep Learning CNNs, RNNs, LLMs, RAG, Prompt Engineering

Data Science Pandas, NumPy, Matplotlib, Seaborn, Power BI

Core CS DSA, OOP, DBMS, Operating Systems, Computer Networks

Full-Stack Development React.js, Next.js, Node.js, Express.js, REST APIs

Databases MongoDB, PostgreSQL, Firebase, Firestore, SQLite

System Design Microservices, Message Queues (Kafka/RabbitMQ), Caching (Redis), Load Balancing

DevOps & Cloud Docker, Kubernetes, AWS (EC2, S3), Git, Vercel, Linux

Developer Tools VS Code, Postman, Jupyter, Git, GitHub

Experience

AI/ML Research Intern

Self-Initiated Research Projects

2024 – Present

- Designed **optimized Transformer architectures** reducing computational complexity from $O(N^2)$ to $O(N \cdot w)$ through restricted self-attention, improving inference speed by **40%** on long-sequence tasks.
- Developed a **multimodal species recognition system** processing image, audio, and text inputs, achieving **92% classification accuracy** across 500+ species using PyTorch and Hugging Face.
- Built an **autonomous human-following robot** with sensor fusion and embedded C, reducing navigation errors by **60%**.

Full-Stack Developer

Coding Contest Platform Project

2025

- Architected a **scalable coding assessment platform** supporting 500+ concurrent users with **99.9% up-time**.
- Designed **RESTful APIs** with JWT authentication, reducing response time by **35%** through query optimization.
- Implemented **real-time leaderboards** and automated code evaluation, improving user engagement by **45%**.

Projects

Efficient Transformer with Restricted Self-Attention

<https://github.com/JNXBAE/coderhub-connect>

- Engineered a **novel attention mechanism** reducing computational complexity from $O(N^2)$ to $O(N \cdot w)$ while preserving performance.
- Achieved **35% faster inference** and **40% lower memory usage** on 10K+ token sequences.
- Published research findings at **IIT Madras** technical symposium, receiving **3rd Prize**.

Tech Stack: PyTorch, CUDA, Hugging Face, NumPy

Coding Contest Platform

<https://github.com/JNXBAE/coding-contest-platform>

- Built **production-ready full-stack platform** with 500+ monthly active users.
- Designed **MongoDB schemas** and optimized queries, reducing API latency by **35%**.
- Implemented **role-based access control** (RBAC) for admins, participants, and evaluators.

Tech Stack: React.js, Node.js, MongoDB, JWT

Multimodal Species Recognition System

<https://github.com/JNXBAE/species-recognition>

- Developed **AI-powered classification system** achieving **92% accuracy** on 500+ species.
- Implemented **feature fusion** combining vision, audio, and text modalities.
- Deployed as **Docker container** with REST API for easy integration.

Tech Stack: PyTorch, OpenCV, Librosa, Docker

Achievements

- **2nd Prize** - National Paper Presentation, MRK College of Engineering (AI-based Prescription Validation System)
- **3rd Prize** - National Paper Presentation, IIT Madras (AI-based Industrial Ventilation System)
- **Top 10%** - Presented research at 5+ inter-college technical symposiums

Certifications

- **NPTEL** – Cloud Computing (Elite certification)
- **IBM** – Getting Started with Artificial Intelligence
- **Great Learning** – UI/UX Design

Additional Information

- **Languages:** English (Fluent), Tamil (Native)
- **Interests:** Open Source, AI Research, Competitive Programming
- **Community:** Active contributor to AI/ML open-source projects